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Topic : NumPy



Numpy

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PYTHON



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What is NumPy ?

- NumPy Stands For Numerical Python.
- NumPy is a Python library used for working with array
- NumPy is the fundamental package for scientific computing in Python.
- It also has functions for working in domain of linear algebra, Fourier transform, and matrices.
- NumPy was created in 2005 by Travis Oliphant. It is an open source project and you can use it freely.

Why Use NumPy ?

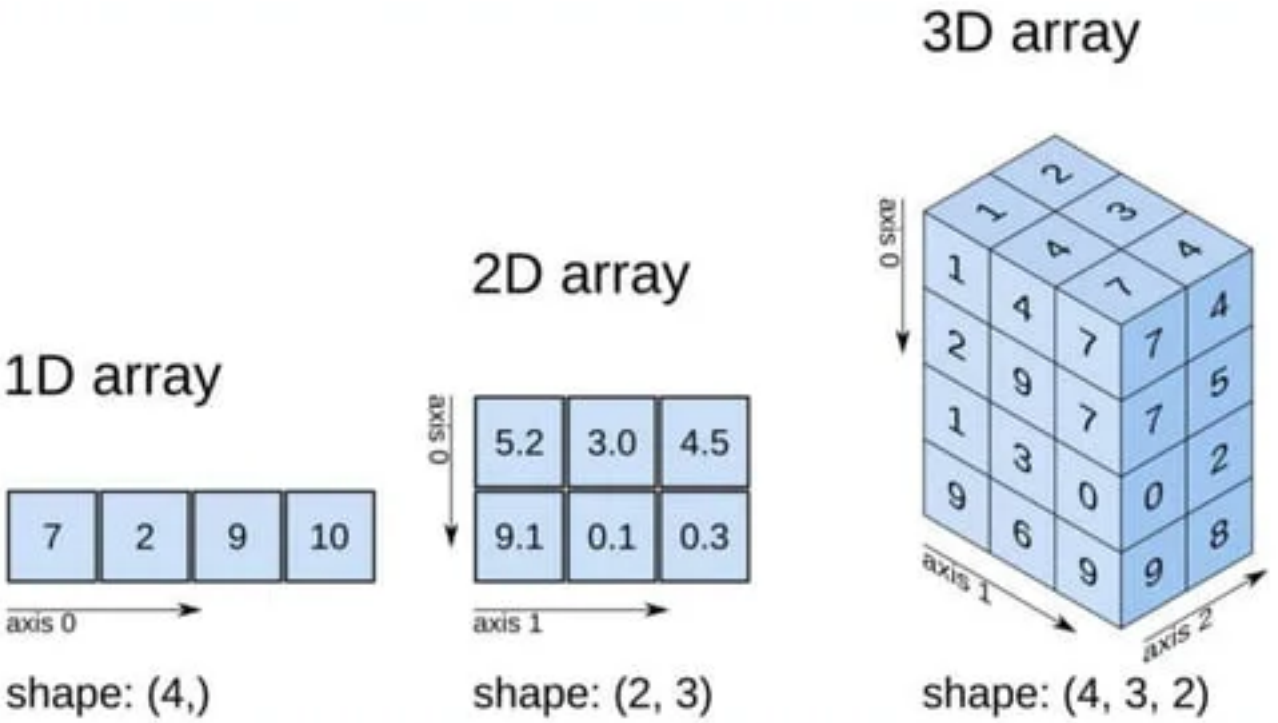
- In Python we have lists that serve the purpose of arrays, but they are slow to process.
- NumPy aims to provide an array object that is up to 50x faster than traditional Python lists.
- The Array object in NumPy is called **ndarray**, it provides a lot of supporting functions that make working with **ndarray** very easy.
- Arrays are very frequently used in data science, where speed and resources are very important.

Why is NumPy Faster Than Lists?

- NumPy arrays are stored at one continuous place in memory unlike lists, so processes can access and manipulate them very efficiently.
- This behavior is called locality of reference in computer science.
- This is the main reason why NumPy is faster than lists. Also it is optimized to work with latest CPU architectures.

Arrays in NumPy

- An Array is a collection of values, organized in a specific order.
- NumPy's main object is the homogeneous multidimensional array.
- It is a table of elements (usually numbers), all of the same type, indexed by a tuple of positive integers.
- In NumPy dimensions are called *axis*. The number of axis is *rank*.
- NumPy's array class is called ndarray. It is also known by the alias array.


1D array
7
2
9
10
axis 0
shape: (4,)
2D array
axis 0
5.2
3.0
4.5
9.1
0.1
0.3
axis 1
shape: (2, 3)
3D array
axis 0
axis 1
axis 2
shape: (4, 3, 2)

1-D Array

- An array that has 0-D arrays as its elements is called uni-dimensional or 1-D array . These are the most common and basic arrays

- Import numpy as np

```
arr = np.array([1, 2, 3, 4, 5])
print(arr)
```

2-D Array

- An array that has 1-D arrays as its elements is called a 2-D array.These are often used to represent matrix or 2nd order tensors.

- Import numpy as np

```
arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr)
```

3-D Arrays

- An array that has 2-D arrays (matrices) as its elements is called 3-D array.These are often used to represent a 3rd order tensor.

- Import numpy as np

```
arr = np.array([[[1, 2, 3], [4, 5, 6]], [[1, 2, 3], [4, 5, 6]]])
print(arr)
```

INITIALIZTION

np.array([1,2,3])	1d array
np.array([(1,2,3),(4,5,6)])	2d array
np.arrange(start,stop,step)	range array
np.linespace(0,2,9)	Add evenly spaced values btw interval to array of length
np.zeros((1,2))	Create and array filled with zeros
np.ones((1,2))	Creates an array filled with ones
np.random.random((5,5))	Creates random array
np.empty((2,2))	Creates an empty array

Array Properties

Syntax	Description
array.shape	Dimensions (Rows,Columns)
len(array)	Length of Array
array.ndim	Number of Array Dimensions
array.size	Number of Array Elements
array.dtype	Data Type
type(array)	Type of Array

Copying & Shorting

np.copy(array)	Creates copy of array
array.sort()	Sorts an array
array.sort(axis=0)	Sorts axis of array

Operations

Operator	Description
np.add(x,y) x + y	Addition
np.substract(x,y) x –y	Subtraction
np.divide(x,y) x / y	Division
np.multiply(x,y) x *y	Multiplication
np.sqrt(x)	Square Root
np.sin(x)	Element-wise sine
np.cos(x)	Element-wise cosine
np.log(x)	Element-wise natural log
np.dot(x,y)	Dot product
np.roots([1,0,-4])	Roots of agiven polynomial coefficients

Thank You...

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